DO NOW — RECALL

Backtracking in computational solutions

Name: _____ Class: _____ Date: _____

1. When does a recursive call stop?

2. What is a search tree node?

3. Why may pruning reduce runtime?

4. Insert 12 into BST root10 with right child15 and left child5. Show the comparisons and position.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

When a base case is met.

2. Model answer

A partial choice/state considered in the search.

3. Model answer

It avoids exploring states proven unable to succeed.

4. Model answer

12>10 ⇒ go right to15; 12<15 ⇒ insert as left child of15.

By the end, you can…

• Make a decision

• Reject a dead end

• Backtrack

Vocabulary — know these terms

backtracking — Explore choices and undo a failed path.

pruning — Discard an impossible branch.

search tree — Branches of possible decisions.

base case — Condition ending a recursive branch.

NEW LEARNING

Read, annotate and explain

Make a decision

At each position choose the current item or skip it. Track index and running total.

Reject a dead end

When the sum exceeds the target (non-negative inputs) or the list ends, undo the choice.

Backtrack

Try the other branch; success means a valid selection has been found. Exponential worst-case growth remains possible.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

Predict first

The first path over-shoots: 2+3+4=9, so reject that branch.

Explain the method

Undo 4, skip 3, then include 4 → 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain choice, dead end and backtrack using [2,3,4] target6. [6]

Question 2

Write a recursive include/exclude Boolean function for subset sum with non-negative integers, including base cases. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why backtracking is useful for 4 queens but not guaranteed polynomial time. Give one safe-placement rule. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Backtracking can require trying an earlier alternative after a dead end.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.